

Dexter Fernandes

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Lidar Perception Engineer

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map, integrating multimodal LiDAR and radiological data for localisation and mapping in inspection scenarios leveraging the Clearpath Husky development platform.
- Evaluated RTAB-Map (LiDAR-only), LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation prior to real world deployment.
- Validated sensor placements for optimal coverage and calibrated LiDAR extrinsics to the robot base such as the Velodyne VLP 16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes using tf2 and message synchronisation and debugged tf trees and frame connectivity.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulation-only continuation of the dissertation autonomy stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM plus LiDAR+RGB fusion SLAM and navigation.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM in 2D and 3D LiDAR across structured scenarios and implemented autonomous navigation with Nav2.
- Developed ROS2 nodes, launch files and parameterised configurations in C++ and Python and implemented tf2 frame trees with message synchronisation and debugged tf connectivity and transform timing issues.
- Implemented point cloud processing using PCL and Open3D for filtering, segmentation and registration to support mapping and localisation stability.
- Implemented an EKF-based localisation layer using robot_localization to fuse wheel odometry and IMU for smooth odom → base_link tracking when running LiDAR-only SLAM, improving motion continuity for Nav2.
- Containerised the stack with Docker on Linux and managed versioning in Git to ensure reproducible builds and consistent execution across development environments.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Designed and deployed real-time perception pipelines on edge devices using PyTorch and OpenVINO, achieving a 15% F1 score improvement with deterministic 24/7 operation under harsh industrial conditions.

- Refactored legacy monolith code into Dockerised microservices leveraging Apache Kafka, improving fault isolation and increasing MTBF by 35% across deployments.
- Implemented continuous integration and testing workflows for edge inference modules, improving maintainability and deployment speed across hardware targets.
- Documented deployment guardrails, failure modes, and recovery procedures, reducing cognitive load for engineers operating and maintaining fielded systems.
- Led on-site validation and diagnostics on oil-rig deployments, conducting root-cause analysis of model failures and improving overall system stability under environmental stressors.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Built asynchronous inference pipelines using OpenVINO and C++17 to maximise throughput on Intel VPU edge hardware under real-time latency and power constraints.
- Collaborated with hardware, software, and QA teams to integrate AI modules into production systems, mentoring engineers on optimisation and deployment practices to improve delivery consistency.
- Re-architected the perception pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.

SKILLS

- **SLAM:** 3D Feature Extraction, Cartographer, Ceres, Eigen, GLIM, LIO-SAM, Pose graph optimisation, RTAB-Map, Scan Matching
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image
- **Robotics Stack:** Gazebo (Harmonic), Nav2, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy), ROS2 image_transport, tf2
- **Programming & Tooling:** Bash, C++17, CMake, colcon, FastAPI, Git, Linux, NumPy, Pandas, PostgreSQL, Python
- **Cloud, Edge & MLOps:** Apache Kafka, CI/CD, Docker, DVC (Data Version Control), Microsoft Azure, MLFlow
- **Deep Learning and Optimisation:** OpenVINO, Optuna, PyTorch, TensorFlow, TensorRT
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership

LiDAR Perception Engineer with extensive experience optimising RTAB-Map and the Velodyne VLP-16 for challenging feature sparse and dense indoor environments and a recent graduate from the University of Bristol. Experienced AI Vision engineer in working with clients such as British Petroleum and Austin Industries building scalable, critical safety solutions under tight latency and power constraints. Proven ability to collaborate effectively in cross-functional teams, communicate complex technical concepts clearly, and thrive in fast-paced environments.